

Making the Hadamard test scale

36× fewer gates with AQC, a phase trap that would have broken it,
and a way to certify the result from its own garbage

Team 8 — Garbage Collectors

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Qiskit Hackathon 2026 — Topic 5

after Faehrman, Eisert & Kueng, *PRL* **135**, 150603 (2025)

14 August 2026

Outline

The question

The instrument

Against the baselines that already exist

AQC: the compressor and its certificate

On real hardware

What we claim

A tag like [R042] is not decoration — it is a **row number** in our public evidence ledger, RESULTS.md: one row per reported number, each with the exact command that produced it. Full repo via the QR code on the last slide. Where a baseline beats us, the slide says so.

The result, up front

36×

fewer two-qubit gates for a Hadamard test's controlled evolution (AQC, $n=7$) — as a *compilation result* [R046]

2 QPUs

same verdict: the gate-count win does **not** reach hardware at $n=6$ — 0.026/0.033 measured vs 0.315/0.401 *pre-registered* [R054]

1 gate

fixes the trap we found: a phase-blind compiler feeding a phase interferometer, $|\Delta\chi|$ **1.33** → **0.008** [R046]

$\equiv 0$

a free error bar: $\langle Q \rangle_W - \langle Q \rangle_U$ is exactly zero by conservation, so its measured value is pure device error — no reference needed [R044]

Track B **completed** past the scaffold (4×10^{-15} validation), benchmarked against the literature's own baselines — which **beat us** on cost, and we say so — and extended to 2D, Fermi–Hubbard and real-device routing. [R042, R043, R049–R052]

Where this sits: the five tracks in the challenge

track	status	evidence
A symmetry-resolved spectral analyser	done	[R012, R013, R019, R022]
B dynamics A/B tester (<i>this deck</i>)	done + extended	[R034, R042, R046, R054]
C eigenstate detective	done	[R032]
D Krylov energy solver from the same records	done	[R016]
E our own proposal: a decoherence witness	done	[R033]

Track A is the mandatory Part B deliverable and is complete: all four populated levels, every symmetry label correct, **12/12 independent seeds**. C, D and E are implemented and validated.

This deck is Track B — the one we took furthest: completed past the scaffold, benchmarked against the literature's own methods, run on two QPUs, and extended to AQC compilation, a 2D lattice and the Fermi–Hubbard model.

The question

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What we claim

You compiled U into something cheaper. Is it still right?

Every practical quantum simulation replaces the target evolution $U(t)$ with a cheaper approximation W — a Trotter product formula, a transpiler setting, or an **AQC-compressed ansatz**.

The question that decides whether the result means anything:

how wrong is W , and wrong in *what*?

Classically you answer it by simulating both. That is exactly the thing you cannot do at the sizes where you need the answer.

Track B answers it *on the device*: put W on one ancilla branch and U on the other, and the interference is the overlap of two dynamics.

W = what you can afford to run

U = what you meant to run

$$\chi_{AB}(t) = \langle \psi | W^\dagger U | \psi \rangle$$

measured, not simulated

The question

The instrument

Against the baselines that already exist

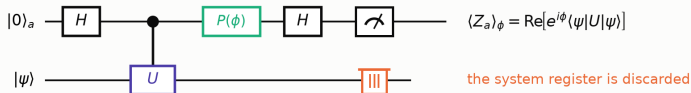
AQC: the compressor and its certificate

On real hardware

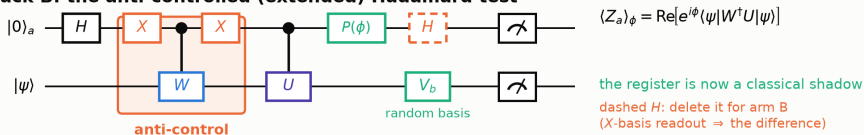
What we claim

Anti-control: how the Hadamard test is extended

Standard Hadamard test



Track B: the anti-controlled (extended) Hadamard test



Why the two X gates:

$$X_a(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes W)X_a = |1\rangle\langle 1| \otimes I + |0\rangle\langle 0| \otimes W$$

Conjugating by X swaps which ancilla state fires the gate, so W acts on the $|0\rangle$ branch and U on the $|1\rangle$ branch. Nothing else changes.

What the ancilla then sees:

$$\frac{1}{\sqrt{2}}(|0\rangle W|\psi\rangle + e^{i\phi}|1\rangle U|\psi\rangle)$$

Two dynamics interfering on one ancilla. $W = I$ gives back the standard test exactly, so this is a strict generalisation.

The only structural change is the X - cW - X sandwich: it makes W fire on the $|0\rangle$ branch, so the ancilla interferes **two dynamics** rather than one. Algebra in appendix A1–A2. [construction after [1] Apps. B & E]

Track B: anti-controlled Hadamard test

Replace controlled- U with “ W when the ancilla is $|0\rangle$, U when it is $|1\rangle$ ”. Then, with the *same* quadrature convention as the ordinary test,

$$\langle Z_a \rangle_\phi = \text{Re} \left[e^{i\phi} \langle \psi | W^\dagger U | \psi \rangle \right], \quad \phi = 0 \rightarrow \text{Re } \chi_{AB}, \quad \phi = -\frac{\pi}{2} \rightarrow \text{Im } \chi_{AB}$$

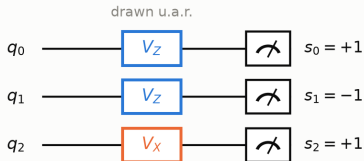
Setting $W = \mathbb{1}$ recovers the ordinary Hadamard test exactly, so this is a strict generalisation — which also means it needs *its own* validation, not inherited trust.

statevector identity, frozen 3-site (3 times $\times$ 2 reps $\times$ 2 quadratures)	8.99×10^{-15} [R034]
statevector identity, 2-site (4 times $\times$ 2 reps $\times$ 2 quadratures)	4.56×10^{-15} [R042]
$\chi_{AB}(0) = 1$ exactly, as it must be (both branches are the identity)	yes

A bug this caught. Our first reference took the $|1\rangle$ -control block as the *upper half* of the gate matrix. The notebook builds the gate with the ancilla as its *lowest* bit, so the block is the odd sublattice. The wrong version passes at $t = 0$ — where both branches are the identity — and fails everywhere else. It was caught before any QPU time was spent.

First, what a classical shadow is

1. Each shot draws its own random basis



$b_j \sim \text{Unif}\{X, Y, Z\}$, independently on every qubit and every shot

All that is stored per shot: $(b_0 b_1 b_2, s_0 s_1 s_2)$

$$\hat{\rho} = \bigotimes_j \left(3 V_{b_j}^\dagger |s_j\rangle \langle s_j| V_{b_j} - 1 \right), \quad \mathbb{E}[\hat{\rho}] = \rho$$

one unbiased snapshot — meaningless alone, exact in expectation once averaged

2. One record, every Pauli at once

$$\hat{\rho} = 3^{w(P)} \prod_{j \in \text{supp}(P)} s_j \mathbf{1}[b_j = P_j]$$

shot	basis b	outcome s	$\widehat{Z_0 Z_1}$	$\widehat{X_0 X_1}$
1	Z Z X	+ - +	-9	0
2	X X Z	+ + -	0	+9
3	Z Y Y	- + +	0	0
4	Z Z Z	+ + +	+9	0
5	Y X Z	- - +	0	0

$w = 2$: a shot lands with probability 3^{-2} and the 3^2 weight cancels it exactly, so $\mathbb{E}[\hat{\rho}] = \text{Tr}[\hat{\rho} \rho]$

q_2 's basis never enters $\widehat{Z_0 Z_1}$: only $\text{supp}(P)$ must match
 $\Rightarrow 3^{-w}$, not 3^{-n} . The price is $\mathbb{E}[\hat{\rho}^2] = 3^w$.

In Track B this is applied to the *system* register — the one a standard Hadamard test throws away. It costs **no new circuits**, and it makes one record set an unbiased estimator for **every** Pauli at once.

(Schematic uses 3 qubits; our Track B instance has 2.) [Huang, Kueng & Preskill, *Nat. Phys.* **16**, 1050 (2020); proof in A4]

What the shadows add: delete the final Hadamard

Before the final Hadamard the state is $(|0\rangle W|\psi\rangle + e^{i\phi}|1\rangle U|\psi\rangle)/\sqrt{2}$. Tracing the ancilla out of the *un-Hadamarded* circuit gives two different things at once:

$$\mathrm{tr}_a[\rho_{\mathrm{out}}] = \frac{1}{2}(W\rho W^\dagger + U\rho U^\dagger), \quad \mathrm{tr}_a[(Z \otimes \mathbb{1})\rho_{\mathrm{out}}] = \frac{1}{2}(W\rho W^\dagger - U\rho U^\dagger)$$

Both are *independent of ϕ* , so one quadrature suffices. Add and subtract:

$\langle O \rangle_W$ and $\langle O \rangle_U$ separately, for every Pauli O , from one family of circuits.

That is the *per-observable disagreement profile* — the route is indicated in [1] Apps. B & E and flagged by the challenge scaffold; the implementation, the sum/difference estimators and their validation — 4.44×10^{-15} across 4 times $\times$ 6 observables, before submission — are ours. [R042]

The ancilla says *how much* W is wrong. The garbage says *which observable*.

The question

The instrument

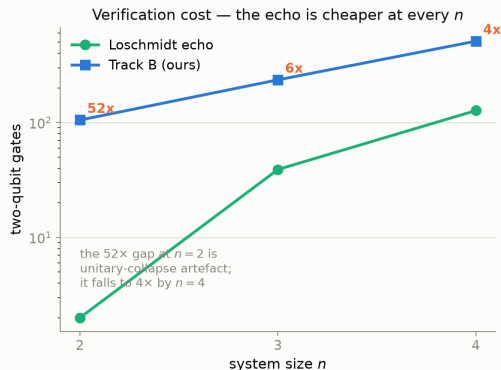
Against the baselines that already exist

AQC: the compressor and its certificate

On real hardware

What we claim

We are not the only way to do this — and not the cheapest



What each method actually returns

gives a number	yes	yes	yes	yes	yes
runs on a QPU	--	yes	yes	yes	yes
keeps the phase	--	--	--	yes	yes
says WHICH observable broke	--	--	--	--	yes
	classical fidelity	Loschmidt echo	Hilbert-Schmidt	Track B ancilla only	Track B + shadows

Loschmidt echo: prepare, apply U , apply $W^\dagger$, un-prepare, read the all-zeros probability.

Hilbert–Schmidt test (Khatri *et al.*, *Quantum* 3, 140 (2019)): Bell pairs on $2n$ qubits give the *state-independent* $|\text{Tr}[W^\dagger U]/d|^2$. Both verified against exact values before being used as baselines.

[R043]

The honest scoreboard

	qubits	2q gates	phase?	state-indep.?	which observable?
classical fidelity (AQC's own score)	—	—	n/a	yes	no
Loschmidt echo	n	2	no	no	no
Hilbert–Schmidt test	$2n$	16	no	yes	no
Track B, ancilla only	$n+1$	136	yes	no	no
Track B + shadows (ours)	$n+1$	136	yes	no	yes

- **The echo is cheaper than us at every n we measured** — $6.0\times$ at $n=3$ and $4.0\times$ at $n=4$. If all you want is a scalar “how close is W to U ”, *use the echo*. [R043]
- The $52\times$ gap at $n=2$ is a *unitary-collapse artefact* (any product of 2-qubit unitaries fuses into one). We quote $4\text{--}6\times$, never $52\times$.
- The HST answers a *stronger* question than either overlap, and costs $2n$ qubits for it.

Our arm has to be justified by what it returns, never by being cheaper.

The question

The instrument

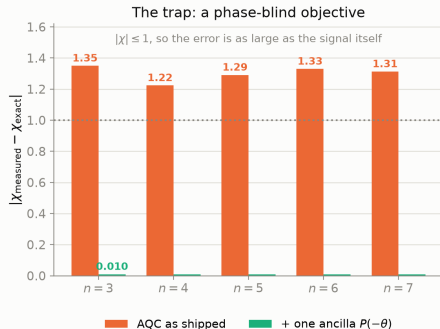
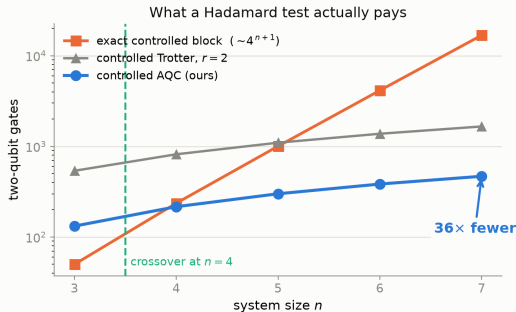
Against the baselines that already exist

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What we claim

AQC makes it scale — crossover at $n = 4$, $36\times$ by $n = 7$



Left. The controlled evolution is the bottleneck: the standard construction builds $|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes U$ as one $(n+1)$ -qubit UnitaryGate, costing $\sim 4^{n+1}$. Controlled AQC grows *roughly linearly*, so the crossover is structural — measured, not extrapolated. **Right.** The trap, next slide. [R046]

2D lattice, and real connectivity. On a 2×4 grid the ideal-count advantage falls only $122.8\times \rightarrow 100.8\times$ at $n=8$. And these counts assume all-to-all: re-transpiled onto `ibm_marrakesh`'s heavy-hex map the gap **widens** — the dense exact block pays $\sim 2.1\times$ for routing, AQC's shallow ansatz only $\sim 1.5\times$, so $n=6$ goes $10.8\times \rightarrow 15.4\times$ and the 2D $n=4$ case **flips from a $0.92\times$ loss to a $1.28\times$ win**. [R049, R052]

The trap: a phase-blind compiler breaks a phase interferometer

qiskit-addon-aqc-tensor maximises **state fidelity** — a *magnitude*, blind to global phase. A Hadamard test measures **exactly that phase**. So a compression converging beautifully (state infidelity 3×10^{-4}) returns $W|\psi\rangle = e^{i\theta} U|\psi\rangle$ with $\theta \approx 2.3\text{--}3.0$ rad, i.e. $|\Delta\chi| \approx 1.3$ on a quantity bounded by $|\chi| \leq 1$ — an error as large as the signal.

Why we nearly missed it. Our earlier certificate reported $|\chi_{AB}|$ — also a magnitude. A magnitude cannot detect a phase error. [R035, now closed by R046]

And it is not only a compiler problem. Re-reading our teammates' `ibm_miami` run (a **Nighthawk-class** square-lattice device — 218 couplers in its own coupling map): $|\chi|$ survival **0.988** — near-perfect-looking — while the *complex* χ is **22.6 σ off**, a 0.77 rad phase error. A magnitude-only figure of merit is structurally blind to what a Hadamard test measures. [R053, R031]

The fix costs one single-qubit gate. $\theta = \arg\langle U\psi|W\psi\rangle$ is free at compile time, and a $P(-\theta)$ **on the ancilla** cancels it exactly: $|\Delta\chi|$ **1.33** $\rightarrow$ **0.008**.

Scope, not buried. (i) AQC is compressed against *one* input state (process infidelity ~ 0.96) — legitimate for a Hadamard test, illegitimate the moment anything feeds it another. [R046] (ii) On **2D Fermi–Hubbard** the gate win survives (34.7 $\times$) but accuracy degrades to the shot-noise floor: cheap, not yet usable as-is. [R051]

The question

The instrument

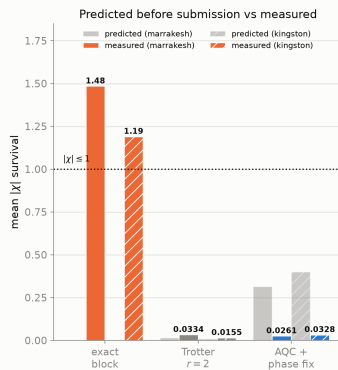
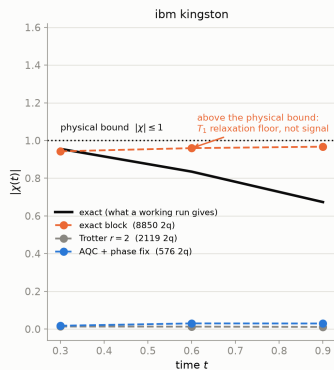
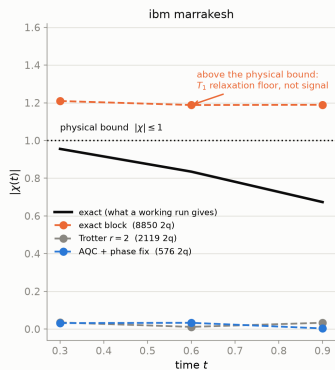
Against the baselines that already exist

AQC: the compressor and its certificate

On real hardware

What we claim

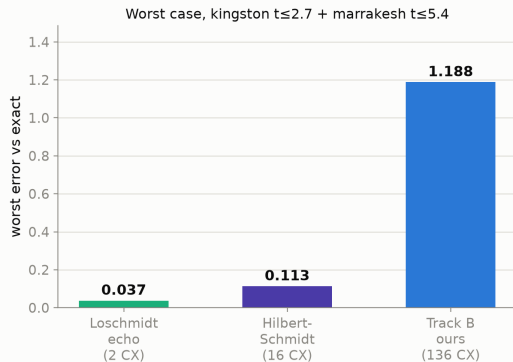
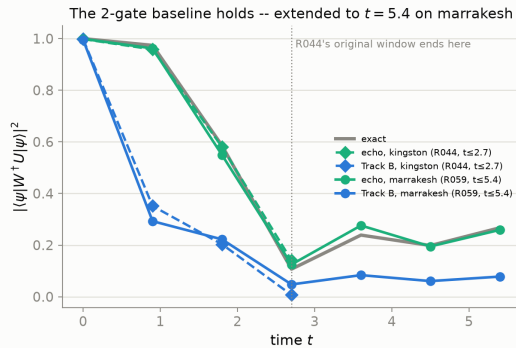
We ran it on a QPU. It failed — and the failure is instructive



$n=6$, 144,000 shots across both devices. **Every arm on both QPUs returned noise** — AQC (576 gates) at **0.026/0.033** vs the *pre-registered* 0.315/0.401. **Two independent QPUs, same verdict.** The orange arm is the lesson: its “survival” **1.485** is **unphysical** ($|\chi| \approx 1.21 > 1$, near constant, quadratures pinned at $+0.85$) — the ancilla sitting in $|0\rangle$ ($602\mu\text{s}$ vs $T_1 = 160\mu\text{s}$). **A magnitude-only metric scores a dead circuit as the winner:** the compiler’s trap, now in our own experiment. [R054]

Why $n=6$, not the headline’s $n=7$? The dead control arm dominates QPU time ($\sim 4\times$ more at $n=7$) for no added contrast — full arithmetic in appendix A7. [R055]

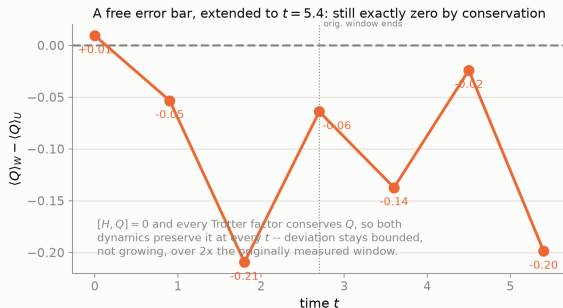
First Track B on a QPU — and the cheap baseline beat us



116 circuits $\times$ 1000 shots, all four arms in *one* job per device, so each comparison is same-device same-hour. Predictions were recorded before submission. **The 136-CX Track-B arm degrades fast** — clean at $t = 0$ (errors ≤ 0.053), noise dominated by $t = 0.9$ — and on kingston it goes wrong in *phase*, not just magnitude ($-0.03 - 0.59i$ against exact $+0.99 + 0.02i$ at $t = 0.9$), so this is not simple damping. marrakesh extended to $t=5.4$ ($2\times$ the original window, *same* 136-gate depth at every t):

the echo's lead does not close — worst-case gap widens from $27\times$ to $32\times$. [R044, R059]

The garbage also ships a free error bar



$Q = \sum_j Z_j$ commutes with H , and every Trotter factor conserves it — so $\langle Q \rangle_W - \langle Q \rangle_U$ is **exactly zero** for any W in this family.

Extended to $t=5.4$ on marrakesh ($2 \times$ R044's original window, one self-consistent job): +0.01, −0.05, −0.21, −0.06, −0.14, −0.02, −0.20.

[R059]

That deviation is **pure device error, detected with no reference state, no exact diagonalisation and no simulation** — **bounded, not growing**, over the longer window.

It is a diagnostic, not a correction: we do *not* post-select on it, which would bias the shadow estimators.

The question

The instrument

Against the baselines that already exist

AQC: the compressor and its certificate

On real hardware

What we claim

What we claim, and what we do not

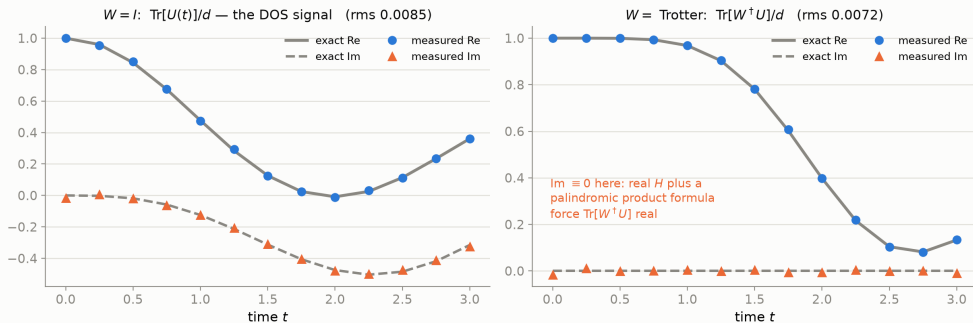
We claim

- **AQC makes the Hadamard test scale as a compilation result:** crossover at $n=4$, $36\times$ fewer two-qubit gates at $n=7$, against the honest baseline and surviving real device routing. [R046, R052]
- **We found and fixed a trap that silently destroys the answer** — a phase-blind objective feeding a phase interferometer — for the cost of one single-qubit gate. [R046]
- Track B **completed**: the per-observable disagreement profile via the X-basis readout, which our own earlier run left undone, validated to 4×10^{-15} . [R042]
- Track B run on a **QPU for the first time here**, against two literature baselines in the same job. [R044]

We do not claim

- **That the gate-count win becomes a hardware win.** We tested it on two QPUs and it did not: every arm returned noise, AQC at 0.026/0.033 against predicted 0.315/0.401. [R054]
- **That AQC helps at small n .** Below the crossover it is worse — $2.6\times$ at $n=3$ — and the compressed circuit is valid **only for the state it was compiled against**. [R046]
- **That our arm is the best way to get the scalar overlap.** It is not — a 2-gate Loschmidt echo is cheaper and, on hardware, $32\times$ more accurate, holding over a $2\times$ -extended time window. [R043, R044, R059]
- Any advantage over classical computation. These are gate counts, not runtimes; the benchmark is small *on purpose*, so every number is graded against truth.

Future work: the same circuit is a density-of-states estimator



Feed Track B the **maximally mixed** state instead of $|\psi\rangle$ and the ancilla measures $\text{Tr}[W^\dagger U]/d$. With $W = I$ that is $\text{Tr}[U(t)]/d$ — the generating function whose Fourier transform is the **density of states**, i.e. Eq. (7) of Goh & Koczor (arXiv:2407.03414v2). So their DOS protocol is the $W=I$ special case of our Track B circuit, not a separate experiment. Verified to 2.2×10^{-15} ; measured on the simulator at rms 0.0085 and 0.0072. [R045]

Reproduce it

- `evidence/scripts/track_b_ab_tester.py` — Track B derivation + statevector validation + simulator study [R034]
- `evidence/scripts/track_b_baselines.py` — Loschmidt echo, Hilbert–Schmidt test, $n=2, 3, 4$ cost scaling [R043]
- `evidence/scripts/track_b_hardware_submit.py` — validate-then-submit, all four arms [R042]
- `evidence/scripts/track_b_hardware_fetch.py` — the analysis behind every hardware number here [R044]
- `evidence/scripts/aqc_compression.py` — AQC + Track-B certification [R035]



github.com/mnsh0409/Qiskit-Hackathon-2026

(full repo; this deck is `deck/trackb.pdf`)

Team 8 — Garbage Collectors

Chao Hsien (M1) · Jiwan Kang (M2) · Ng Siu Hin (M3) · Su Wei-Chang (M4) · Tina Tien (M5)

Thank you — questions welcome.

References

Papers

- [1] P. K. Faehrmann, J. Eisert, R. Kueng, *In the Shadow of the Hadamard Test: Using the Garbage State for Good and Further Modifications*, Phys. Rev. Lett. **135**, 150603 (2025); arXiv:2505.15913. — the protocol; the anti-controlled test and X-basis readout (Apps. B & E).
- [2] H.-Y. Huang, R. Kueng, J. Preskill, *Predicting many properties of a quantum system from very few measurements*, Nat. Phys. **16**, 1050 (2020). — classical shadows and the 3^w estimator.
- [3] S. Khatiri, R. LaRose, A. Poremba, L. Cincio, A. T. Sornborger, P. J. Coles, *Quantum-assisted quantum compiling*, Quantum **3**, 140 (2019). — the Hilbert–Schmidt test baseline.
- [4] Goh & Koczor, arXiv:2407.03414. — fermionic density of states (the future-work link, slide “Future work”).

Software Qiskit 2.5.1 and qiskit-ibm-runtime 0.49 (IBM); qiskit-addon-aqc-tensor 0.3.1 (the AQC compiler); quimb [J. Gray, JOSS **3**(29), 819 (2018)] + JAX (MPS backend).

Benchmark & tracks the hackathon Topic-5 participant notebook defines the XXZ benchmark and Tracks A–E; all figures are our own work generated by scripts in the repository (the companion master deck additionally reproduces Fig. 1 of [1] under CC BY 4.0, attributed on that slide).

Summary — what we contributed, and what we found

Built & verified

- All **five** challenge tracks; the mandatory analyser at **12/12 seeds**, 56/56 notebook checkpoints [R019]
- **Track B completed** past the scaffold: per-observable $\langle O \rangle_W$ vs $\langle O \rangle_U$ readout, validated to 4.4×10^{-15} [R042]
- Benchmarked vs the field's own baselines — **which beat us on cost**, and we say so [R043, R044]
- **AQC crossover measured**: $n=4$, **36×** fewer gates by $n=7$; survives 2D, Fermi–Hubbard, real routing [R046, R049, R051, R052]
- 58-row public evidence ledger; bug log incl. our retractions; everything reproducible from the repo

Found & learned

- **The phase trap**: a magnitude-blind compiler breaks a phase interferometer — $|\Delta\chi|$ **1.33** → **0.008** with **one ancilla gate** [R046]
- The same magnitude-hides-phase failure, **three independent times** [R046, R053, R054]
- **Pre-registered falsification**: the gate-count win does *not* reach today's QPUs (0.026/0.033 vs 0.315/0.401, two devices); visibility needs $\epsilon_{\text{eff}} \lesssim 2 \times 10^{-3}$ [R054, R055]
- A **reference-free error bar** from conservation, measured on hardware [R044]; the DOS estimator as Track B's $W=I$ special case [R045]
- Where SKQD works and why not here: the localisation boundary, in Δ/J and U/t [R047, R050]

Real wins and real negatives, every number sourced — that is the contribution.

Appendix — derivations

A1 anti-control A2 the Track B identity A3 the two readouts

A4 classical shadows A5 shadows on the Track B circuit A7 the $n=6/n=7$
arithmetic

A1 — Anti-control, and A2 — the Track B identity

A1. A controlled gate fires on $|1\rangle$. Conjugating the ancilla by X swaps the two branches:

$$X_a \left(|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes W \right) X_a = |1\rangle\langle 1| \otimes \mathbb{1} + |0\rangle\langle 0| \otimes W$$

so $X - cW - X$ applies W exactly when the ancilla is $|0\rangle$. Composing it with an ordinary cU puts a *different* evolution on each branch.

A2. Propagating $|0\rangle_a |\psi\rangle$ through the circuit:

$$\begin{aligned} & \xrightarrow{H} \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |\psi\rangle \xrightarrow{X_c W X} \frac{1}{\sqrt{2}} (|0\rangle W |\psi\rangle + |1\rangle |\psi\rangle) \xrightarrow{cU} \frac{1}{\sqrt{2}} (|0\rangle W |\psi\rangle + |1\rangle U |\psi\rangle) \\ & \xrightarrow{P(\phi)} \frac{1}{\sqrt{2}} (|0\rangle W |\psi\rangle + e^{i\phi} |1\rangle U |\psi\rangle) \xrightarrow{H} \frac{1}{2} \left[|0\rangle (W + e^{i\phi} U) |\psi\rangle + |1\rangle (W - e^{i\phi} U) |\psi\rangle \right] \end{aligned}$$

Using $\|(W \pm e^{i\phi} U)\psi\|^2 = 2 \pm 2 \operatorname{Re}[e^{i\phi} \langle \psi | W^\dagger U | \psi \rangle]$,

$$\langle Z_a \rangle_\phi = \frac{1}{4} \left(\|(W + e^{i\phi} U)\psi\|^2 - \|(W - e^{i\phi} U)\psi\|^2 \right) = \operatorname{Re}[e^{i\phi} \langle \psi | W^\dagger U | \psi \rangle]$$

$W = \mathbb{1}$ returns the standard test, so this is a strict generalisation. [verified numerically to 4.56×10^{-15} , R042]

A3 — the two readouts: why deleting one gate gives the difference

(Route indicated in [1] Apps. B & E; derivation and numerical validation below are ours, R042.)

Stop *before* the final Hadamard. With $|\Phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle W|\psi\rangle + e^{i\phi}|1\rangle U|\psi\rangle)$ and $\rho = |\psi\rangle\langle\psi|$,

$$\rho_{\text{out}} = |\Phi\rangle\langle\Phi| = \frac{1}{2} \left[|0\rangle\langle 0| \otimes W\rho W^\dagger + |1\rangle\langle 1| \otimes U\rho U^\dagger + e^{-i\phi} |0\rangle\langle 1| \otimes W\rho U^\dagger + \text{h.c.} \right]$$

The off-diagonal ancilla blocks are killed by both partial traces below, so *both results are independent of ϕ* — one quadrature suffices:

$$\text{tr}_a[\rho_{\text{out}}] = \frac{1}{2}(W\rho W^\dagger + U\rho U^\dagger) \equiv S, \quad \text{tr}_a[(Z \otimes \mathbb{1})\rho_{\text{out}}] = \frac{1}{2}(W\rho W^\dagger - U\rho U^\dagger) \equiv D$$

Hence for *any* observable O , from the same shots,

$$\langle O \rangle_W = \text{Tr}[O(S + D)], \quad \langle O \rangle_U = \text{Tr}[O(S - D)]$$

Keeping the final H gives the overlap; deleting it gives the per-observable profile. [verified]

to 4.44×10^{-15} over 4 times $\times$ 6 observables, R042]

A4 — classical shadows: the inversion and its variance

Measure each qubit in a basis b_j drawn uniformly from $\{X, Y, Z\}$, with $V_b W V_b^\dagger = Z$. The single-qubit measurement channel is

$$\mathcal{M}_1(\sigma) = \mathbb{E}_{b,s} \left[\langle s | V_b \sigma V_b^\dagger | s \rangle V_b^\dagger | s \rangle \langle s | V_b \right] = \frac{1}{3} \sigma + \frac{1}{3} \text{Tr}(\sigma) \mathbb{1}$$

— expand σ over $\{\mathbb{1}, X, Y, Z\}$: the identity part always survives, and each traceless part survives only when *its own* basis is drawn, with probability $1/3$. Inverting on the traceless part,

$$\mathcal{M}_1^{-1}(A) = 3A - \text{Tr}(A) \mathbb{1} \quad \implies \quad \hat{\rho} = \bigotimes_j \left(3 V_{b_j}^\dagger |s_j\rangle \langle s_j| V_{b_j} - \mathbb{1} \right), \quad \mathbb{E}[\hat{\rho}] = \rho$$

For a Pauli string P , the single-qubit traces are 1 ($P_j = \mathbb{1}$), $3s_j$ ($P_j = b_j$), and 0 otherwise, so

$$\hat{P} = 3^{w(P)} \prod_{j \in \text{supp}(P)} s_j \mathbb{1}[b_j = P_j], \quad \mathbb{E}[\hat{P}^2] = 3^{2w} \cdot \underbrace{3^{-w}}_{\text{all } w \text{ match}} = 3^{w(P)}$$

The second moment is **exactly** 3^w , independent of ρ — which is why low-weight observables are cheap and weight- n ones are not. [HKP, *Nat. Phys.* **16**, 1050 (2020); CONVENTIONS §(Pauli-string estimator)]

A5 — putting them together on the Track B circuit

Each shot of arm B records $(b_0 \dots b_{n-1}, s_0 \dots s_{n-1}, a)$ with $a = \pm 1$ the ancilla outcome. Form $\hat{P}$ from the system record as in A4. Then, by A3,

$$\mathbb{E}[\hat{P}] = \text{Tr}[P S] = \frac{1}{2}(\langle P \rangle_W + \langle P \rangle_U), \quad \mathbb{E}[a \hat{P}] = \text{Tr}[P D] = \frac{1}{2}(\langle P \rangle_W - \langle P \rangle_U)$$

so a single sample mean of each gives both dynamics separately, for every P at once:

$$\widehat{\langle P \rangle_W} = \overline{\hat{P}} + \overline{a \hat{P}}, \quad \widehat{\langle P \rangle_U} = \overline{\hat{P}} - \overline{a \hat{P}}$$

Two consequences we use on the slides.

- Any $O = \sum_P c_P P$ follows by linearity — H , Q , and the hopping term all come from the same records.
- If $[O, H] = 0$ and every factor of W conserves O , then $\langle O \rangle_W - \langle O \rangle_U \equiv 0$ *identically*. Its measured value is therefore **pure device error**, with no reference needed — the free error bar on slide 21. [R044]

A6 — Track B → density of states, and where the phase matters

With $\rho = \mathbb{1}/d$ the Track B identity of A2 becomes state-*independent*:

$$\chi_{AB}(t) = \text{Tr}[W^\dagger U \rho] \xrightarrow{\rho=\mathbb{1}/d} \frac{1}{d} \text{Tr}[W^\dagger U] \quad \text{and at } W = \mathbb{1} : \quad \frac{1}{d} \text{Tr}[U(t)]$$

which is Eq. (7) of Goh & Koczor up to normalisation. Since $\text{Tr}[A] = \sum_z \langle z|A|z\rangle$, the mixed input is realised by enumerating the d computational basis states — the deterministic version of their Statement 2 (a 1-design, e.g. random $\{\mathbb{1}, X\}^{\otimes n}$, suffices).

Where the phase is worth having — measured, not assumed:

	$\max \text{Im} $	does the HST lose information?
$W = \mathbb{1}$ (the DOS signal)	0.504	yes — it returns magnitude only
$W = \text{2nd-order Suzuki (Trotter error)}$	2.8×10^{-14}	no — it is real anyway

Real H gives $U^* = U^\dagger$, and a *palindromic* product formula gives $W^* = W^\dagger$, which forces $\text{Tr}[W^\dagger U]$ real. So our advantage over the HST is the full complex value *for the DOS case*, and only the register size ($n+1$ vs $2n$) for the Trotter case. **We checked this rather than claiming the phase everywhere.**

A measured nonzero Im in the second row is therefore another reference-free error signal. [R045]

A7 — the $n=6$ vs $n=7$ arithmetic, in full

Measured anatomy at $n=6$ (marrakesh: cz 68 ns, $T_1=160\ \mu\text{s}$, $T_2=74\ \mu\text{s}$) [R054]

arm	2q	dur	t/T_2	meas.
exact	8850	$602\ \mu\text{s}$	8.1	1.49[†]
Trotter	2119	$144\ \mu\text{s}$	1.9	0.033
AQC	576	$39\ \mu\text{s}$	0.53	0.026

[†]unphysical: T_1 floor, not signal

Effective error — a *lower bound*, not a point estimate:

$$\epsilon_{\text{eff}} \geq \frac{-\ln S}{N_{2q}} = \frac{-\ln 0.026}{576} = 6.3 \times 10^{-3}$$

(kingston: 5.9×10^{-3} ; bound because measured $|\chi|$ sits near the estimator's noise floor $\sigma\sqrt{\pi/2} \approx 0.020$ at these shots) — $\geq 3\text{--}4\times$ the calibrated $1.7\text{--}2.0 \times 10^{-3}$.
Budgeted: 0.315 (gate error) $\times e^{-39.2/74} = 0.59$ (ancilla T_2) = 0.186 ; measured $0.026 \Rightarrow$ residual $7.2\times$ unattributed; visibility ($S \gtrsim 0.3$) needs $\epsilon_{\text{eff}} \leq 2.1 \times 10^{-3}$.

The same job at $n=7$, transpiled — not estimated [R055]

arm	2q	dur	pred.
exact	36,114	2.46 ms	3×10^{-28}
Trotter	2,549	$173\ \mu\text{s}$	6×10^{-3}
AQC	702	$48\ \mu\text{s}$	0.245*

*gate-error only; at the *measured* ϵ_{eff} : $e^{-6.1 \times 10^{-3} \times 702} \approx 0.013$

What moving to $n=7$ does, quantitatively:

- ratio (compilation): $16,812/468 = \mathbf{35.9\times}$ — *grows*
- AQC signal: $0.026 \rightarrow \sim 0.013$ — *halves*
- quantum time: whole job $\times 3.4$, exact control arm $\times 4.1$ ($36,114/8,850$)

Doubles the headline, halves the signal, quadruples the bill. The win becomes measurable when

$\epsilon_{\text{eff}} \lesssim 2 \times 10^{-3}$ (gap $\geq 3\times$) — a device generation (Nighthawk's target regime), not a larger n . [R046, R052,

R054, R055]